

ORLAND, Norway

WMO: 012410

Lat: 63.699N

Lon: 9.604E

Elev: 9

StdP: 101.22

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-10.0	-7.9	-14.5	1.1	-7.8	-12.6	1.3	-6.0	19.1	2.8	17.5	2.5	6.6	130	0.765

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.8	23.6	16.5	21.5	15.5	19.7	14.5	17.4	22.0	16.2	20.1	15.2	18.5	4.7	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.3	10.9	19.2	14.4	10.2	17.7	13.6	9.7	16.6	48.9	22.0	45.3	20.1	42.4	18.6	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 16.0	14.0	12.4	DB	-12.9	27.0	2.4	2.4	-14.7	28.8	-16.1	30.2	-17.4	31.5	-19.2	33.3	(4)
(5)			WB	-13.5	19.3	2.4	1.7	-15.2	20.5	-16.6	21.5	-17.9	22.4	-19.7	23.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.7	0.7	0.4	1.7	5.4	8.7	11.4	14.1	14.1	11.5	7.1	3.4	1.5
	DBStd	6.01	3.60	3.97	3.42	3.18	3.45	2.93	2.99	2.84	2.78	3.09	3.60	4.06
	HDD10.0	1618	289	267	257	144	68	16	1	1	16	98	198	264
	HDD18.3	4257	548	501	515	389	299	209	138	134	206	348	448	522
	CDD10.0	414	0	0	0	4	27	58	127	129	61	8	1	0
	CDD18.3	11	0	0	0	0	1	1	6	4	0	0	0	0
	CDH23.3	72	0	0	0	0	2	6	41	23	0	0	0	0
	CDH26.7	9	0	0	0	0	0	0	7	2	0	0	0	0
Wind	WSAvg	5.9	7.9	7.6	6.8	5.4	4.8	4.4	3.8	4.0	5.3	6.3	6.9	7.4
Precipitation	PrecAvg	1013	85	75	69	57	50	67	74	87	119	116	95	113
	PrecMax	1486	223	205	175	128	102	192	137	178	310	241	222	275
	PrecMin	617	3	6	10	14	16	15	13	23	28	22	5	32
	PrecStd	178	51	40	38	28	20	33	32	39	57	54	52	54
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	8.2	10.7	17.5	22.9	24.2	27.5	25.9	20.6	16.0	11.9	9.8
		MCWB	5.9	5.9	6.0	10.4	14.2	15.1	19.0	17.5	14.4	11.8	8.9	8.0
	2%	DB	7.1	7.1	8.3	14.2	19.6	21.0	23.9	23.2	18.6	14.0	10.1	8.5
		MCWB	5.4	5.3	5.2	8.6	12.5	14.0	17.2	16.6	13.5	10.5	7.8	6.9
	5%	DB	6.1	6.2	7.2	11.9	16.6	18.7	21.6	21.0	16.9	12.3	9.0	7.4
		MCWB	4.5	4.5	4.7	7.5	11.1	13.3	16.2	15.6	12.9	9.5	7.1	5.8
	10%	DB	5.2	5.3	6.2	9.9	14.2	16.7	19.4	19.0	15.5	11.1	8.0	6.4
		MCWB	3.6	3.6	4.2	6.5	9.7	12.5	15.1	14.6	12.1	8.7	6.0	4.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.8	7.0	7.1	11.0	14.7	16.3	20.1	18.5	15.5	12.4	9.9	8.7
		MCDB	7.3	7.5	8.5	16.3	21.3	21.1	26.4	24.2	18.8	15.0	11.1	9.3
	2%	WB	5.9	6.0	6.1	9.1	12.9	14.7	17.8	17.3	14.2	11.1	8.5	7.3
		MCDB	6.7	6.7	7.4	13.3	18.0	19.3	22.6	21.9	17.0	13.1	9.5	8.0
	5%	WB	4.8	4.9	5.2	7.9	11.5	13.7	16.6	16.1	13.5	10.1	7.4	6.1
		MCDB	5.7	5.9	6.8	10.7	15.9	17.5	20.6	19.9	15.9	11.9	8.7	7.1
	10%	WB	3.8	3.8	4.4	7.0	10.1	12.6	15.4	15.1	12.7	9.2	6.3	5.0
		MCDB	5.0	5.0	5.9	9.1	13.2	15.9	18.9	18.1	15.1	10.8	7.6	6.2
Mean Daily Temperature Range	5% DB	MDBR	3.8	4.1	4.9	6.1	6.6	6.0	6.7	6.8	5.7	4.5	3.8	3.9
		MCDBR	3.9	4.4	5.6	9.0	11.0	10.4	11.4	10.7	7.7	5.4	4.4	4.4
		MCWBR	3.6	3.7	3.8	4.7	5.4	5.3	5.9	5.1	4.0	3.4	3.6	4.0
	5% WB	MCDBR	3.8	4.0	5.1	7.8	10.0	9.3	10.6	9.5	6.8	4.9	4.4	4.3
		MCWBR	3.9	3.9	3.9	4.5	5.3	5.3	6.0	5.0	4.0	3.6	4.1	4.2
Clear-Sky Solar Irradiance	taub	N/A	0.306	0.333	0.356	0.354	0.356	0.364	0.352	0.337	0.312	0.363	0.359	
	taud	N/A	2.232	2.304	2.369	2.441	2.468	2.472	2.502	2.522	2.416	2.598	2.514	
	Ebn at Noon	N/A	610	755	823	860	864	845	824	762	615	328	161	
	Edh at Noon	N/A	59	82	96	97	97	94	84	68	49	26	16	
All-Sky Solar Radiation	RadAvg	0.13	0.60	1.82	3.48	4.64	5.00	4.65	3.57	2.11	0.92	0.22	0.06	
	RadStd	0.02	0.10	0.26	0.29	0.24	0.48	0.42	0.36	0.26	0.13	0.04	0.01	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.70	N/A	N/A	N/A	N/A	N/A	-170	-251	+84	N/A
(47) Regional (36 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-143	-189	+67	N/A

Nomenclature: See separate page